

# Welding-Robot — Standard Operating Procedure

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## Welding-Robot — Standard Operating Procedure

**Audience:** Maintenance Tech · Line Manager · Quality Worker  
**Applies to:** Line-C, station 2, Welding-Robot

### 1. Purpose

The Welding-Robot performs MIG / TIG robotic welds joining the components delivered by the loader. It is the second-highest fault contributor on Line-C (after Leak-Test) and a key driver of weld-related rejects at Weld-Inspection.

### 2. Normal operating envelope

- Ideal cycle time: **25 s**
- Max acceptable cycle time: **32 s**
- Expected reject rate: **≤ 3 %**
- Fault probability per cycle: **0.7 %**
- Input buffer capacity: **5 parts**

### 3. Idle reasons

| idle_reason | Meaning here                                               | First check                    |
|-------------|------------------------------------------------------------|--------------------------------|
| Starved     | Component-Loader (station 1) is not feeding parts.         | Verify Component-Loader state. |
| Blocked     | Weld-Inspection (station 3) cannot accept the welded part. | Verify Weld-Inspection state.  |

### 4. Faults & corrective actions

#### 4.1 Wire-Feed-Jam

- **Symptoms in telemetry:** fault\_type = "Wire-Feed-Jam". Arc fails mid- cycle; Weld-Inspection (next station) flags missing welds.
- **Likely root cause:** wire snagged in the liner, drive rolls worn, or

spool brake too tight.

- **Corrective action:**
  1. LOTO. Open the wire-feed drive.
  2. Clear the snagged wire; trim and reload through the liner.
  3. Inspect drive rolls; replace if grooved.
- **Restart criteria:** three test welds pass visual and CT contour check.

## 4.2 Tip-Wear

- **Symptoms in telemetry:** fault\_type = "Tip-Wear". Erratic arc start; porosity rejects flagged at Weld-Inspection.
- **Likely root cause:** contact tip bore worn beyond service limit.
- **Corrective action:**
  1. LOTO. Replace the contact tip; inspect the gas diffuser.
  2. Verify wire stick-out per recipe.
- **Restart criteria:** five welds with stable arc current trace.

## 4.3 Gas-Flow-Low

- **Symptoms in telemetry:** fault\_type = "Gas-Flow-Low". Porosity defects cluster on Weld-Inspection.
- **Likely root cause:** shielding-gas regulator drifted, hose leak, or cylinder near-empty.
- **Corrective action:**
  1. Verify cylinder pressure and flow at the torch.
  2. Leak-check hose connections with soapy water.
  3. Replace regulator if it cannot hold setpoint.
- **Restart criteria:** flow stable at 15–20 L/min for 5 minutes.

# 5. Preventive maintenance schedule

- **Daily:** check wire-feed tension; gas-flow readback.
- **Weekly:** replace contact tip; clean nozzle.
- **Monthly:** robot path verification; cable management check.
- **Quarterly:** robot axis backlash measurement.

# 6. References

- Maintenance\_Workflow\_Overview.pdf
- Safety\_Lockout\_Tagout\_Procedure.pdf
- Line-C\_01\_Component-Loader\_SOP.pdf
- Line-C\_03\_Weld-Inspection\_NDT.pdf